



Intratumoral and Peritumoral Habitat Radiomics of MRI predicts pathologic complete response to neoadjuvant chemoimmunotherapy in oral squamous cell carcinoma

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Abstract:

Background: Neoadjuvant chemoimmunotherapy (NACI) shows promise in oral squamous cell carcinoma (OSCC), but reliable noninvasive biomarkers for predicting pathologic complete response (pCR) remain scarce. Radiomics integrating intratumoral and peritumoral heterogeneity across multi-sequence MRI may offer novel insights into treatment response evaluation.

Methods: The data of 212 patients with OSCC after NACI were retrospectively collected and analyzed. Among these patients, 56 (26.4%) achieved pCR after NACI. Intratumoral and peritumoral habitat imaging (HI) was achieved using the K-means clustering algorithm applied to T1-weighted imaging (T1WI), axial T2-weighted imaging with fat suppression (T2WI), and contrast-enhanced T1-weighted imaging with fat suppression (T1C). Moreover, intratumoral and peritumoral HI models were

constructed and compared using the receiver operating characteristic curve (ROC). Five-fold cross-validation was performed to mitigate model overfitting.

Results: Intratumoral HI models derived from different sequences and the Intratumoral Fusion model exhibited favorable predictive ability, with AUCs of 0.738-0.817 and 0.729-0.789 in the training and testing cohorts, respectively. Moreover, peritumoral HI models displayed marginally higher predictive abilities compared to intratumoral HI and Fusion models, with AUCs of 0.734-0.869 and 0.788-0.802 in the training and testing cohorts, respectively. Meanwhile, the decision model with peritumoral habitat features (PHFs), intratumoral habitat features (IHF), and 3 clinical features displayed the highest performance, with average AUCs of 0.913 and 0.843 in both respective cohorts. Among the most important features screened by SHAP, three IHFs and one PHF could effectively distinguish between the lower and higher groups of programmed cell death ligand 1 (PD-L1) Combined Positive Score ($t = 2.027-2.275$, $p < 0.05$), whilst two PHFs were highly correlated with CD45+ white blood cell densities in the stroma ($r = 0.958, -0.920$, $p < 0.05$), which were associated with pCR.

Conclusions: Integrated intratumoral and peritumoral HI derived from multi-sequence MRI offers a high predictive capacity for pCR following NACI in OSCC patients.

Key Words: Oral squamous cell carcinoma, Radiomics, Intratumoral and peritumoral, Habitat imaging, Pathologic complete response

Highlights

In a retrospective analysis of 212 OSCC patients undergoing neoadjuvant chemoimmunotherapy, multi-sequence MRI-derived intratumoral and peritumoral habitat imaging demonstrated robust predictive value for pCR.

The integrated decision model combining peritumoral habitat features (PHFs), intratumoral habitat features (IHF), and clinical characteristics achieved superior predictive performance (AUC 0.913/0.843 in training/testing cohorts), outperforming single-sequence or single-region models.

SHAP-based visualization identified critical habitat features were associated with PD-L1 CPS and CD45+ in the stroma which was correlated with pCR, providing biological interpretability for personalized treatment optimization.

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Introduction

As is well documented, oral squamous cell carcinoma (OSCC) is a relatively prevalent malignant tumor among head and neck squamous cell carcinoma (HNSCC)¹ and is characterized by a poor prognosis. To date, its treatment is largely similar to that of HNSCC. The predominant treatment strategy for HNSCC patients is a comprehensive regimen integrating surgery, radiotherapy, chemotherapy, and targeted therapy. However, a substantial proportion of patients, approximately 40%-60%, ultimately experience local recurrence or distant metastasis, with a 5-year survival rate below 50%². In recent years, immunotherapy, especially immune checkpoint inhibitors (ICIs), has emerged as a promising approach for the treatment of OSCC. Although neoadjuvant chemoimmunotherapy (NACI) can significantly enhance the pathologic complete response (pCR) rate in patients with locally advanced HNSCC, including those with OSCC, roughly 45% of patients fail to achieve pCR³. Consequently, there is an urgent and unmet need for the early prediction of NACI efficacy in OSCC patients.

Radiomics, a method for quantifying intratumoral heterogeneity (ITH), has been extensively utilized in diverse aspects of HNSCC⁴⁻¹⁴. In the specific subfield of OSCC, earlier studies have also focused on its value in predicting staging¹⁵, cervical lymph node metastasis¹⁶⁻²⁰, local recurrence²¹, bone invasion²², and overall survival (OS)²³. Regarding the prediction of treatment response in OSCC after NACI, only Liu et al.¹⁴ attempt to predict post-treatment pCR in HNSCC patients by extracting histogram-related features within tumors from multi-sequence MRI. Nonetheless, it is worth acknowledging that their sample size was limited. Similarly, Lin et al.¹² developed a radiomics model based on multi-sequence MRI to predict post-treatment pCR in HNSCC patients, albeit using the traditional radiomics approach. Traditional radiomics typically assumes a homogeneous distribution of

features within the region of interest (ROI), thereby overlooking intratumoral regional disparities and facing challenges in capturing the complex spatial distribution information within tumors.

In contrast, sub-region-based habitat imaging (HI) principally combines quantitative image analysis with the underlying tumor pathophysiology. Constructing a "habitat map" of the tumor may reveal internal heterogeneity, thereby mirroring tumor growth, drug resistance, pathological grade, and prognosis²⁴⁻²⁸. Moreover, given that it can more effectively quantify sub-regions that are more relevant to tumor growth or invasiveness, it is superior to a simple whole-tumor analysis. Concurrently, numerous studies have explored the application of peritumoral radiomics features to characterize the microenvironment for predicting patient outcomes^{29,30}. However, these efforts have been predominantly limited to the overall analysis of the peritumoral region, with several studies demonstrating that peritumoral HI can also contribute to enhancing the predictive capacity of the model³¹⁻³³.

To the best of our knowledge, no studies have employed intratumoral and peritumoral HI to predict pCR following NACI in OSCC. Therefore, this study aimed to establish intratumoral and peritumoral HI models based on multi-sequence MRI to predict the pCR of OSCC patients after NACI. In addition, an interpretability analysis method was introduced to enhance the biological interpretability of the model. To ensure transparency and reproducibility in radiomics analysis, this study adheres to the Transparency In The Reporting of Artificial Intelligence (TITAN) guideline³⁴.

Methods

Patients

This retrospective study was approved by the institutional review board (IRB) of XXX (A) and XXX (B). Due to the retrospective design and use of de-identified data,

individual patient consent was waived by the IRB. All patient data, including clinical and imaging information, were anonymized before analysis, with personal identifiers removed and sensitive data encrypted. This study comprehensively reviewed the clinical information of OSCC patients who underwent NACI at two institutions from March 2020 to December 2023. Initially, a total of 244 patients, consisting of 204 patients from Institution A and 40 from Institution B, were identified. Participants were eligible for inclusion if they met the following criteria: (a) histopathologically confirmed OSCC at the initial diagnosis; (b) No treatment received prior to the first MRI examination; (c) Receipt of NACI; (d) Underwent MRI examination after NACI; (e) Underwent surgical intervention. The exclusion criteria were as follows: (a) Absence of clinical data and postoperative pathological results; (b) MRI image artifacts significantly affecting image quality. Ultimately, the dataset comprised 187 patients from Institute A and 25 patients from Institute B. Among them, 56 achieved pCR, whereas the remaining 156 patients were classified as non-pCR (Figure 1). This study had been reported in line with the REMARK criteria³⁵.

For each patient, baseline characteristics, such as gender, age, smoking history, drinking history, diagnosis of tumor occurrence, pre-treatment TNM staging, baseline hematological index, HPV status, and Programmed Death-Ligand 1 (PD-L1) Combined Positive Score (CPS) were collected from medical records. The number of immunotherapy cycles and TNM staging after neoadjuvant therapy were also collected for analysis. It should be noted that only 33 patients underwent PD-L1 CPS assessment.

MR examination

In the present study, all patients underwent head and neck MRI scanning. Specifically, Institution A employed the Siemens 1.5T MRI scanner (Aera), the General Electric (GE) 1.5T MRI scanner (optima MR360), and the United Imaging

3T MRI scanner (uMR780), while Institution B utilized the GE 1.5T MRI scanner (Signa HDXT) and the United Imaging 3T MRI scanner (uMR790). Subsequently, axial T1-weighted imaging (T1WI), axial fat-suppressed T2-weighted imaging (T2WI), and axial contrast-enhanced T1-weighted imaging (T1C) were acquired for all patients. These highly standardized imaging sequences play a crucial role in the MRI staging process of OSCC. Among them, T1WI provides excellent anatomical detail, T2WI highlights tissue heterogeneity, and T1C emphasizes tumor angiogenesis. Detailed imaging parameters are presented in Table S1.

Intratumoral and Peritumoral Region Generation

Referencing the corresponding T1WI and T2WI images, ROI encompassing the whole tumor across all layers was delineated on T1C images by a radiologist with 10 years of experience. In order to ensure the accuracy of the contour, the contoured area was reviewed by another radiologist with 15 years of experience. Regions of necrosis, hemorrhage, or edema were excluded from the ROIs. Delineation was performed using open-source Slicer software (<http://www.slicer.org>). Next, the Onekey v.2.2.3platform (<http://medai.icu/>) was utilized to expand the tumor boundary outward by 3 mm, which was defined as the peritumoral region¹². The radiologist with 15 years of experience reviewed the peritumoral region to ensure that the expanded regions excluded surrounding cavities and bone tissues, thereby improving the accuracy and reliability of peritumoral segmentation. The workflow for radiomics analysis is illustrated in Figure 2.

Image Preprocessing

To minimize the impact of images from different imaging devices on the generalizability of radiomics models, this study conducted a two-step post-processing approach. First, the N4 bias field correction method based on the SimpleITK toolkit in Python 3.7.1 was initially employed to eliminate the influence of uneven signal

intensity caused by inhomogeneous magnetic field variations across images. Subsequently, the zoom toolkit was utilized to resample these images from three sequences, adjusting the voxel size to $1\text{mm} \times 1\text{mm} \times 1\text{mm}$. Trilinear interpolation was adopted for image resampling, while the nearest neighbor interpolation technique was applied to the masks. Finally, the T1WI and T2WI images were registered onto the T1C images, thus providing an accurate and standardized image dataset for subsequent image analysis.

Habitat Imaging and Radiomics Feature Extraction

Intratumoral and peritumoral habitat regions were primarily defined through the utilization of the `sklearn.cluster.Kmeans` toolkit in Python 3.7.1. Compared to the Gaussian Mixture Model and the OTSU Method, the K-means clustering method boasts several advantages, including high computational efficiency, robust adaptability to data distribution, and enhanced interpretability of results³⁶⁻³⁸. In this study, HI was conducted on T1WI, T2WI, and T1C images. Each voxel within the intratumoral and peritumoral areas encompassed paired data of these three sequences. The combination of these paired data from all voxels across the entire patient cohort generated a data array with a dimension of $N \times 3$, where N denotes the number of image voxels, and 3 corresponds to the parameters of the three sequence images. Thereafter, a K-means clustering method was constructed based on the aforementioned dataset to allocate each voxel to a specific cluster or habitat. K-means clustering is an iterative approach that partitions n data points into k clusters by assigning each data point to the cluster with the closest mean. This method initiates with a random assignment, recalculates the means, and iterates until the cluster assignments stabilize or a predefined limit is attained. The Sum of Squared Errors (SSE) was employed to assess clustering quality. SSE serves as a crucial evaluation metric in the K-means clustering algorithm, estimating the sum of the squared

distances from each data point to the center of its affiliated cluster. The elbow method was utilized to determine the optimal value of k by identifying the inflection point of the SSE, where the rate of decline abruptly slows. In this study, the inflection points for both the intratumoral and peritumoral regions were 5, thus subdividing the sub-regions into h1, h2, h3, h4, and h5. Meanwhile, we also performed K-means clustering analysis on data from both centers to ensure the robustness of the K-means clustering algorithm across different cohorts, and found that the optimal number of clusters was consistently 5 for both datasets (Figure S1).

In order to avoid overfitting as much as possible, 107 low-dimensional radiomics features were extracted from each sub-region using the Pyradiomics package of Python (version 3.7.1), including the shape ($n=14$), first order ($n = 18$), gray-level co-occurrence matrix (GLCM, $n = 24$), gray-level run-length matrix (GLRLM, $n = 16$), gray-level size zone matrix (GLSZM, $n = 16$), neighborhood gray-tone difference matrix (NGTDM, $n = 5$), and gray-level dependence matrix (GLDM, $n = 14$). Finally, 1070 radiomics features were acquired from each sequence for subsequent model construction and analysis.

Feature Selection and Model Construction

This study employed a two-stage feature selection pipeline to optimize model performance. First, radiomics features were standardized using the Z-score approach to ensure that the mean value of the features was 0 and the variance was 1. Dimensionality reduction was then performed using Pearson correlation coefficient. When the correlation coefficient $|R|$ between two features exceeded 0.95, one of the features was eliminated. The remaining features underwent further refinement, which was performed using the 10-fold cross-validated LassoCV method to identify the most relevant ones for the model.

In this study, five machine learning algorithms including Logistic Regression (LR), Support Vector Machine (SVM), K-Nearest Neighbors (KNN), Decision Tree (DT) and Random Forest (RF), were used to construct predictive models. To evaluate the prediction models' validity, five-fold cross-validation was also performed on the dataset. In comparison to the bootstrapping method, five-fold cross-validation is less susceptible to the randomness of data partitioning, yields relatively consistent evaluation outcomes, and thus effectively mitigates the risk of overfitting³⁹. Meanwhile, model performance was evaluated using average area under the curve (AUC) value of the receiver operating characteristic (ROC) curve across folds. This approach minimized overfitting and ensured robust evaluation.

This study developed five pCR prediction models: Intratumoral HI models, Intratumoral Fusion model, Peritumoral HI models, Peritumoral Fusion model, and the decision model. The Intratumoral and Peritumoral HI models were constructed using HI features from different sequences, while their respective Fusion models integrated HI features from three sequences. The decision model incorporated clinical characteristics along with both intratumoral and peritumoral HI features.

Biological Interpretability Analysis

In order to evaluate the contribution of the finally screened features to the predictive model, SHapley Additive exPlanations (SHAP) analysis was performed to evaluate the interpretability of the linear model, revealing the complex mechanism of machine learning model prediction. SHAP visualizes the contribution of each feature to the model prediction by assigning an importance value to each feature in each prediction, indicating how it increases or decreases the probability of a specific outcome. The SHAP analysis was carried out using the explainers. Linear function of the shap toolkit in Python (version 3.7.1).

To further enhance model interpretability, the association between the important radiomic features after SHAP screening in the decision model and PD-L1CPS was comprehensively explored. Given the correlation between PD-L1CPS and treatment effectiveness in head and neck carcinoma, a threshold of CPS greater than or equal to 15 was applied for subsequent analysis. Meanwhile, CD45+ white blood cells, CD3+ T cells, and CD19+ B cells in baseline biopsy tissue were quantified using multiplex immunofluorescence assays in pCR patients (n=3) and non-pCR patients (n=3) from Institution B to identify biomarkers related to pCR. The cells were separately quantified in the tumor and stroma. By systematically analyzing the correlation between important radiomic features in the final model and the aforementioned biomarkers, this study aimed to expand our understanding of the potential mechanism underlying disease occurrence and development, providing a valuable theoretical reference for clinical diagnosis and treatment.

Statistical Analysis

Statistical analyses were performed using Python version 3.7.1. Regarding clinical baseline characteristics, continuous variables were expressed as mean±standard deviation and compared using the independent-samples t-test or the Mann -Whitney U test. Ordinal variables were presented as median (first quartile, third quartile) and compared using the chi-square test. It is important to note that a few minimal clinical features, such as the baseline hematological index, were missing. These missing values were primarily imputed using the median value. For radiomics features with missing data, the machine learning algorithm KNNImputer was employed for imputation. In the context of pathological validation, the t-test was used to evaluate correlations between the screened radiomics features and PD-L1CPS, with $p < 0.05$ regarded as statistically significant. Pearson correlation analysis was conducted to evaluate the association between peritumoral subregion features and

screened biomarkers, with $|R| > 0.75$ indicating a strong correlation. To evaluate the discriminatory ability of the models, ROC curves, calibration curves, and confusion matrix were utilized, and model performance was evaluated by calculating the AUC, accuracy, recall, precision, and F1 score to comprehensively and objectively evaluate the reliability and validity of this study.

Results

Patient characteristics

The demographic characteristics of the study cohort are listed in Table 1. Among the 212 patients included in the analysis, 56 individuals (26.4 %) achieved pCR. Significant differences were observed between the pCR and non-pCR groups in pre-N status ($p = 0.042$), number of immunotherapy cycles($p = 0.006$), post-T status ($p = 0.006$), PLT ($p = 0.019$)and MON($p = 0.046$).

The comparison between different models

Given that LR algorithm consistently outperformed other machine learning algorithms across different models (Tables S2–S4), this study primarily presents the results derived from LR. Figures 3A and Table 2 summarize the performance of the HI and Fusion models, with all LR algorithms achieving satisfactory results. The peritumoral HI and Fusion models were minimally higher than those of the intratumoral models. Specifically, peritumoral HI models demonstrated average AUCranges of 0.748–0.788 (training) and 0.734–0.790 (testing), while peritumoral Fusion model achieved 0.869/0.822. In comparison, intratumoral HI models showed AUCs of 0.738–0.793 (training) and 0.729–0.774 (testing), and intratumoral Fusion model scored 0.817/0.789. Complementary performance metrics, including precision, recall, and F1 scores, also further supported these trends (Table 2). Specifically, peritumoral HI models demonstrated superior discriminative ability with precision

ranging from 0.777–0.797 (training) and 0.759–0.782 (testing), recall values between 0.944–0.956 (training) and 0.901–0.928 (testing), and F1 scores of 0.855–0.869 (training) and 0.828–0.837 (testing). Intratumoral HI models achieved comparable results, with precision of 0.767–0.802 (training) and 0.752–0.770 (testing), recall of 0.930–0.966 (training) and 0.900–0.968 (testing), and F1 scores of 0.848–0.861 (training) and 0.819–0.852 (testing). Fusion models exhibited consistent improvements across these metrics. Peritumoral Fusion model attained precision/recall/F1 scores of 0.831/0.930/0.877 (training) and 0.783/0.902/0.827 (testing), while intratumoral Fusion model achieved 0.816/0.942/0.874 (training) and 0.803/0.926/0.789 (testing).

An optimal decision model was constructed based on peritumoral habitat features (PHF_S), intratumoral habitat features (IHF_S), and clinical features. The average AUC values of the decision model in the training and testing sets were 0.913 and 0.843, respectively. In terms of precision, recall, and F1 score, the decision model achieved values of 0.884, 0.955, and 0.918 in the training set, whereas in the testing set these values were 0.835, 0.897, and 0.840. Details are presented in Table 3 and Figure 3B. Figure 3C displays the selected features and their corresponding coefficients in this model, consisting of 18 PHF_S, 16 IHF_S, and 3 clinical features (Number of induction cycles, postT, AGR). Meanwhile, Figure 3B delineates the ROC curve of the decision model and the confusion matrix in the 5-fold cross-validation set. The calibration curves for this model in the training and testing sets are presented in Figure S2.

Interpretability analysis

SHAP analysis was conducted to rank the importance of features in the 5-fold cross-validation models to enhance the interpretability of the optimal decision model (Figure 4A). Interestingly, the PHF_S were more dominant, considering that four of the

top-ranked radiomics features in terms of importance were PHFs. Furthermore, one case was randomly selected from each of the pCR and non-pCR groups for illustrative purposes. The results revealed that within the pCR group, IHF was ranked highest in terms of importance (Figure 4B). Conversely, in the non-pCR group, PHF held the top position (as depicted in Figure 4C). Given the relatively comparable importance of certain features (as delineated in Figure S3), the top three features from each individual cross-validation model were selected for subsequent analysis. The specific features are summarized in Table 4.

Further biological interpretability analysis demonstrated that CD45+ white blood cell densities were significantly higher in the stroma of pCR patients compared to non-pCR patients ($P=0.03$) (Figure 5A-H). Moreover, two PHFs (P_T1C_h2_dm_SmallDependenceHighGrayLevelEmphasis, P_T1C_h4_F_Variance) among the aforementioned features were highly correlated with CD45+ white blood cell densities in the stroma ($r=0.958, -0.920, p < 0.05$) (Table 4 and Figure 5I-J). In addition, this study preliminarily explored the relationship between the aforementioned features and PD-L1 CPS. Three IHFs (I_T1_h3_Firstorder_Median, I_T2_h5_Firstorder_10Percentile, I_T2_h5_Firstorder_RootMeanSquared) and one PHF (P_T1C_h5_Firstorder_Skewness) could distinguish between lower and higher CPS scores ($t=2.027\sim2.275, p < 0.05$) (Table 4 and Figure 5K-N).

Discussion

To the best of our knowledge, this is the first study to apply multi-parameter MRI-based intratumoral and peritumoral HI to predict pCR following NACI in OSCC patients. As anticipated, the overall efficacy of the peritumoral HI models was marginally higher than that of the intratumoral HI models. The decision model composed of 18 PHFs, 16 IHFs, and 3 clinical features (Number of induction cycles,

postT, AGR) achieved the highest efficacy. In terms of biological verification, among the 11 most important features screened by SHAP, 3 ITHs and 1 PTHs were associated with PD-L1 CPS, whilst 2 PTHs were highly correlated with CD45+white bloodin the stroma, which was correlated with pCR.

Notably, traditional radiomics methods have been extensively applied in OSCC, yet research on predicting the pCR of NACI for OSCC remains extremely limited. Liu et al.¹⁴ extracted histogram features from pre- and post-treatment MRI images (T1WI, T2WI, T1C) and ADC maps of HNSCC patients. Their models demonstrated promising predictive ability (AUC = 0.83), which is lower than that of our models (AUCs = 0.913 and 0.843 in the training and test sets, respectively). Moreover, the sample size of their study was relatively small, with only 11 out of 24 patients achieving pCR. In another study, Lin et al.¹² integrated intratumoral and peritumoral radiomics features derived from T2WI and T1C sequences to predict pCR in HNSCC, achieving high AUC values (0.904, 0.860 and 0.849 in the internal training, test and external validation sets). However, their final model included PD-L1 CPS, which necessitates invasive testing and expensive cost. In contrast, our model does not directly incorporate biomarkers. Instead, we identified intratumoral and peritumoral habitat radiomics features associated with biomarkers such as PD-L1 CPS and CD45+ cell, and used them to develop the decision model. This approach enables our predictive model to have broader applications.

Furthermore, the integration of ITFs and PTFs into our decision model demonstrated superior predictive performance for pCR in comparison to single intratumoral or peritumoral HI model. It is noteworthy that the peritumoral HI model exhibited a modest advantage over the intratumoral HI model. This result aligns with the findings of previous studies that investigated the prediction of epidermal growth factor receptor (EGFR) mutations³², the invasiveness of lung adenocarcinoma³¹, and

microvascular invasion in hepatocellular carcinoma³³. Intratumoral HI model can elucidate the internal biological behavior of tumors, the integrated model broadens the dimensionality of data by incorporating peritumoral features, enhancing its ability to distinguish between responders and non-responders to immunotherapy. The peritumoral region functions as a critical interface between tumor tissue and normal tissue, contain such as immune cells, blood vessels, fibroblasts, and the extracellular matrix. Research has demonstrated that cells within the peritumoral region, including lymphocytes, dendritic cells, high endothelial venule cells, and myofibroblastic cancer-associated fibroblasts, exert a substantial influence on the immune response to tumor cells⁴⁰. Concurrently, peritumoral blood vessels, induced by factors such as vascular endothelial growth factor (VEGF), influence drug delivery and immune cell trafficking. Consequently, the intricate tumor microenvironment in the peritumoral area is instrumental in determining the efficacy of immunotherapy. These factors facilitate a more comprehensive reflection of the tumor's biological behavior within the peritumoral region, consequently enhancing the performance of predicting pCR. This finding is consistent with the observations reported in previous research^{31,32,33}.

In terms of the biological interpretability of imaging features, this study identified, through SHAP analysis, that two PHFs were highly correlated with CD45+ cells in the stroma, while multiple IHFs were associated with the PD-L1 CPS in tumor cells. CD45, a leukocyte marker, encompasses various immune cell subsets, including T lymphocytes, B lymphocytes, macrophages, dendritic cells, and neutrophils. Our findings indicate that CD45+ cells (i.e., leukocytes) in the peritumoral stroma prior to treatment are associated with pCR in OSCC. Higher CD45+cell density has been demonstrated to be a significant predictor of enhanced immune-tumor interactions and improved responses to immunotherapy in this study. Meanwhile, these PHFs may potentially serve as indicators for monitoring changes in the peritumoral immune

microenvironment, providing a novel perspective for evaluating the efficacy of immunotherapy. Numerous studies had demonstrated a positive correlation between higher levels of immune cells, particularly specific CD8+ T cell subsets, and enhanced survival outcomes^{41,42}. Conversely, the PD-L1 CPS in tumor cells functions as a significant biomarker for predicting the efficacy of immunotherapy in head and neck cancer. Patients with higher CPS scores frequently demonstrate superior immune responses⁴³. Also, there are some studies that investigated the microenvironment of HNSCC through the application of radiomics. For instance, Wang et al.⁴⁴ employed preoperative CT-based radiomics to predict the enrichment status of CD8+ T cells in HNSCC. Katsoulakis et al.⁴ developed a model to predict CD8+ T cell infiltration in HNSCC through radiomic analysis using data from The Cancer Imaging Archive and The Cancer Genome Atlas. In the present study, IHFs were found to be predominant in a randomly selected patient sample with pCR, and multiple IHFs were correlated with CPS. In a randomly selected non-pCR patient sample, PHFs were found to be predominant. It is imperative that future research endeavors prioritize the execution of expansive sample studies, which are essential for further exploration and elucidation of the underlying phenomena.

Nevertheless, this study is subject to several limitations. Firstly, as a retrospective study, there are risks of selection and recall bias. Despite the employment of a medium-sized sample, five-fold cross-validation, and stable radiomics features to reduce overfitting⁴⁵, biases may still exist. Furthermore, the performance of the model may be subject to confounding by multiple clinical covariates, including tumour localisation and chemotherapy strategy. Therefore, a prospective, multi-centre study is required. Secondly, although correlations between tumour microenvironment radiomics features and PD-L1 CPS, CD45+ cells in the stroma were established, further immunological validation at the tissue and molecular level is required. Thirdly,

in order to minimise data consistency discrepancies across different centres, the use of functional MRI image such as diffusion weighted imaging (DWI) was excluded. Subsequent research will investigate multi-sequence combinations to enhance prediction accuracy. Finally, in the absence of a consensus definition of the peritumoral region^{32,33,46}, it is acknowledged that variations in the definition may have a consequential impact on the performance of the models. Further study was needed to conduct an in-depth exploration of the effects of different peritumoral areas on radiomics models, and also investigate the associations between the radiomics features and biomarkers to determine the optimal definition of the peritumoral region.

In summary, the MR-based integrated intratumoral and peritumoral HI models can provide valuable information for predicting the response to NACI in OSCC, and SHAP interpretability analysis enhances the interpretability of the decision model, thereby enabling the formulation of personalized treatment for patients.

Provenance and peer review

Not commissioned, externally peer-reviewed

Conflicts of Interest

The authors declare no conflicts of interest.

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Ethical Approval

The number is [2024]KYLL(SO4)

Consent

The requirement for obtaining informed consent from the patients was waived due to the retrospective design.

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Reference

1. Sung H, Ferlay J, Siegel RL, et al. Global Cancer Statistics 2020: GLOBOCAN Estimates of Incidence and Mortality Worldwide for 36 Cancers in 185 Countries. *CA Cancer J Clin.* 2021;71(3):209-249. doi:10.3322/caac.21660.
2. Gatta G, Botta L, Sánchez MJ, et al. Prognoses and improvement for head and neck cancers diagnosed in Europe in early 2000s: The EURO CARE-5 population-based study. *Eur J Cancer.* 2015;51(15):2130-2143. doi:10.1016/j.ejca.2015.07.043.
3. Wu D, Li Y, Xu P, et al. Neoadjuvant chemo-immunotherapy with camrelizumab plus nab-paclitaxel and cisplatin in resectable locally advanced squamous cell carcinoma of the head and neck: a pilot phase II trial. *Nat Commun.* 2024;15(1):2177. doi:10.1038/s41467-024-46444-z.
4. Katsoulakis E, Yu Y, Apte AP, et al. Radiomic analysis identifies tumor subtypes associated with distinct molecular and microenvironmental factors in head and neck squamous cell carcinoma. *Oral Oncology.* 2020;110:104877. doi:10.1016/j.oraloncology.2020.104877.
5. Wang K, Zhou Z, Wang R, et al. A multi-objective radiomics model for the prediction of locoregional recurrence in head and neck squamous cell cancer. *Med Phys.* 2020;47(10):5392-5400. doi:10.1002/mp.14388.
6. Keek S, Sanduleanu S, Wesseling F, et al. Computed tomography-derived radiomic signature of head and neck squamous cell carcinoma (peri)tumoral tissue for the prediction of locoregional recurrence and distant metastasis after

- concurrent chemo-radiotherapy. *PLoS One*. 2020;15(5):e0232639. doi:10.1371/journal.pone.0232639.
7. Mes SW, van Velden FHP, Peltenburg B, et al. Outcome prediction of head and neck squamous cell carcinoma by MRI radiomic signatures. *Eur Radiol*. 2020;30(11):6311-6321. doi:10.1007/s00330-020-06962-y.
 8. Leger S, Zwanenburg A, Leger K, et al. Comprehensive Analysis of Tumour Sub-Volumes for Radiomic Risk Modelling in Locally Advanced HNSCC. *Cancers (Basel)*. 2020;12(10):3047. doi:10.3390/cancers12103047.
 9. Ou D, Blanchard P, Rosellini S, et al. Predictive and prognostic value of CT based radiomics signature in locally advanced head and neck cancers patients treated with concurrent chemoradiotherapy or bioradiotherapy and its added value to Human Papillomavirus status. *Oral Oncology*. 2017;71:150-155. doi:10.1016/j.oraloncology.2017.06.015.
 10. Salvatore A, Rebecca R, Marco B, et al. Prognostic role of pre-treatment magnetic resonance imaging (MRI)-based radiomic analysis in effectively cured head and neck squamous cell carcinoma (HNSCC) patients. *Acta Oncologica*. 2024;60(9):1192-1200. doi:10.1080/0284186X.2021.1924401.
 11. Yoon H, Ha S, Kwon SJ, et al. Prognostic value of tumor metabolic imaging phenotype by FDG PET radiomics in HNSCC. *Annals of Nuclear Medicine*. 2021;35(3):370-377. doi:10.1007/s12149-021-01586-8.
 12. Lin P, Xie W, Li Y, et al. Intratumoral and peritumoral radiomics of MRIs predicts pathologic complete response to neoadjuvant chemoimmunotherapy in patients with head and neck squamous cell carcinoma. *J Immunother Cancer*. 2024;12(11):e009616. doi:10.1136/jitc-2024-009616.
 13. Yuan Y, Ren J, Shi Y, et al. MRI-based radiomic signature as predictive marker for patients with head and neck squamous cell carcinoma. *Eur J Radiol*. 2019;117:193-198. doi:10.1016/j.ejrad.2019.06.019.
 14. Liu H, Zhu C, Wang X, et al. Prediction of pathological complete response in locally advanced head and neck squamous cell carcinoma treated with neoadjuvant chemo-immunotherapy using volumetric multisequence MRI histogram analysis. *Neuroradiology*. 2024;66(6):919-929. doi:10.1007/s00234-024-03339-6.

15. Li W, Li Y, Gao S, et al. Integrating lipid metabolite analysis with MRI-based transformer and radiomics for early and late stage prediction of oral squamous cell carcinoma. *BMC Cancer*. 2024;24(1):795–12. doi:10.1186/s12885-024-12533-x.
16. Chen Z, Yu Y, Liu S, et al. A deep learning and radiomics fusion model based on contrast-enhanced computer tomography improves preoperative identification of cervical lymph node metastasis of oral squamous cell carcinoma. *Clin Oral Investig*. 2023;28(1):39. doi:10.1007/s00784-023-05423-2.
17. Jin N, Qiao B, Zhao M, et al. Predicting cervical lymph node metastasis in OSCC based on computed tomography imaging genomics. *Cancer Med*. 2023;12(18):19260-19271. doi:10.1002/cam4.6474.
18. Tomita H, Yamashiro T, Heianna J, et al. Nodal-based radiomics analysis for identifying cervical lymph node metastasis at levels I and II in patients with oral squamous cell carcinoma using contrast-enhanced computed tomography. *Eur Radiol*. 2021;31(10):7440-7449. doi:10.1007/s00330-021-07758-4.
19. Wang W, Liang H, Zhang Z, et al. Comparing three-dimensional and two-dimensional deep-learning, radiomics, and fusion models for predicting occult lymph node metastasis in laryngeal squamous cell carcinoma based on CT imaging: a multicentre, retrospective, diagnostic study. *EClinicalMedicine*. 2024;67:102385. doi:10.1016/j.eclinm.2023.102385.
20. Lan T, Kuang S, Liang P, et al. MRI-based deep learning and radiomics for prediction of occult cervical lymph node metastasis and prognosis in early-stage oral and oropharyngeal squamous cell carcinoma: a diagnostic study. *Int J Surg*. 2024;110(8):4648-4659. doi:10.1097/JS9.0000000000001578.
21. Ling X, Alexander GS, Molitoris J, et al. Radiomic biomarkers of locoregional recurrence: prognostic insights from oral cavity squamous cell carcinoma preoperative CT scans. *Front Oncol*. 2024;14:1380599. doi:10.3389/fonc.2024.1380599.
22. Öztürk EMA, Ünsal G, Erişir F, et al. Prediction of bone invasion of oral squamous cell carcinoma using a magnetic resonance imaging-based machine learning model. *Eur Arch Otorhinolaryngol*. 2024;281(12):6585-6597. doi:10.1007/s00405-024-08862-z.

23. Ling X, Alexander GS, Molitoris J, et al. Identification of CT-based non-invasive radiomic biomarkers for overall survival prediction in oral cavity squamous cell carcinoma. *Sci Rep.* 2023;13(1):21774. doi:10.1038/s41598-023-48048-x.
24. Lee DH, Park JE, Kim N, et al. Tumor Habitat Analysis Using Longitudinal Physiological MRI to Predict Tumor Recurrence After Stereotactic Radiosurgery for Brain Metastasis. *Korean J Radiol.* 2023;24(3):235-246. doi:10.3348/kjr.2022.0492.
25. Kim M, Park JE, Yoon SK, et al. Vessel size and perfusion-derived vascular habitat refines prediction of treatment failure to bevacizumab in recurrent glioblastomas: validation in a prospective cohort. *Eur Radiol.* 2023;33(6):4475-4485. doi:10.1007/s00330-022-09164-w.
26. Wu J, Gensheimer MF, Zhang N, et al. Tumor Subregion Evolution-Based Imaging Features to Assess Early Response and Predict Prognosis in Oropharyngeal Cancer. *J Nucl Med.* 2020;61(3):327-336. doi:10.2967/jnumed.119.230037.
27. Xie C, Yang P, Zhang X, et al. Sub-region based radiomics analysis for survival prediction in oesophageal tumours treated by definitive concurrent chemoradiotherapy. *EBioMedicine.* 2019;44:289-297. doi:10.1016/j.ebiom.2019.05.023.
28. Wu J, Cao G, Sun X, et al. Intratumoral Spatial Heterogeneity at Perfusion MR Imaging Predicts Recurrence-free Survival in Locally Advanced Breast Cancer Treated with Neoadjuvant Chemotherapy. *Radiology.* 2018;288(1):26-35. doi:10.1148/radiol.2018172462.
29. Braman N, Prasanna P, Whitney J, et al. Association of Peritumoral Radiomics With Tumor Biology and Pathologic Response to Preoperative Targeted Therapy for HER2 (ERBB2)-Positive Breast Cancer. *JAMA Netw Open.* 2019;2(4):e192561. doi:10.1001/jamanetworkopen.2019.2561.
30. Jiang Y, Wang H, Wu J, et al. Noninvasive imaging evaluation of tumor immune microenvironment to predict outcomes in gastric cancer. *Ann Oncol.* 2020;31(6):760-768. doi:10.1016/j.annonc.2020.03.295.
31. Shang Y, Zeng Y, Luo S, et al. Habitat Imaging With Tumoral and Peritumoral Radiomics for Prediction of Lung Adenocarcinoma Invasiveness on

- Preoperative Chest CT: A Multicenter Study. *AJR Am J Roentgenol.* 2024;223(4):e2431675. doi:10.2214/AJR.24.31675.
32. Shang Y, Chen W, Li G, et al. Computed Tomography-derived intratumoral and peritumoral radiomics in predicting EGFR mutation in lung adenocarcinoma. *Radiol Med.* 2023;128(12):1483-1496. doi:10.1007/s11547-023-01722-6.
 33. Wang C, Wu F, Wang F, et al. The Association Between Tumor Radiomic Analysis and Peritumor Habitat-Derived Radiomic Analysis on Gadoxetate Disodium-Enhanced MRI With Microvascular Invasion in Hepatocellular Carcinoma. *J Magn Reson Imaging.* 2025;61(3):1428-1439. doi:10.1002/jmri.29523.
 34. Agha RA, Mathew G, Rashid R, et al. Transparency In The reporting of Artificial INtelligence – the TITAN guideline. *Premier Journal of Science* 2025;10;100082. doi: 10.70389/PJS.100082.
 35. McShane LM, Altman DG, Sauerbrei W, et al. REporting recommendations for tumour MARKer prognostic studies (REMARK). *Br J Cancer.* 2005;93(4):387-391. doi:10.1038/sj.bjc.6602678.
 36. Dextraze K, Saha A, Kim D, et al. Spatial habitats from multiparametric MR imaging are associated with signaling pathway activities and survival in glioblastoma. *Oncotarget.* 2017;8(68):112992-113001. doi:10.18632/oncotarget.22947.
 37. Ikoma Y, Kishimoto R, Tachibana Y, et al. Reference region extraction by clustering for the pharmacokinetic analysis of dynamic contrast-enhanced MRI in prostate cancer. *Magn Reson Imaging.* 2020;66:185-192. doi:10.1016/j.mri.2019.08.034.
 38. Chen K, Sui C, Wang Z, et al. Habitat radiomics based on CT images to predict survival and immune status in hepatocellular carcinoma, a multi-cohort validation study. *Transl Oncol.* 2025;52:102260. doi:10.1016/j.tranon.2024.102260.
 39. Ron K. A study of cross-validation and bootstrap for accuracy estimation and model selection. 1995.

40. Liu YT, Liu HM, Ren JG, et al. Immune-featured stromal niches associate with response to neoadjuvant immunotherapy in oral squamous cell carcinoma. *Cell Rep Med*. 2025;6(3):102024. doi:10.1016/j.xcrm.2025.102024.
41. Zhao M, Schoenfeld JD, Egloff AM, et al. T cell dynamics with neoadjuvant immunotherapy in head and neck cancer. *Nat Rev Clin Oncol*. 2025;22(2):83-94. doi:10.1038/s41571-024-00969-w.
42. Ren S, Lan T, Wu F, et al. Intratumoral CD103+ CD8+ T cells predict response to neoadjuvant chemoimmunotherapy in advanced head and neck squamous cell carcinoma. *Cancer Commun (Lond)*. 2023;43(10):1143-1163. doi:10.1002/cac2.12480.
43. Burtneß B, Rischin D, Greil R, et al. Pembrolizumab Alone or With Chemotherapy for Recurrent/Metastatic Head and Neck Squamous Cell Carcinoma in KEYNOTE-048: Subgroup Analysis by Programmed Death Ligand-1 Combined Positive Score. *J Clin Oncol*. 2022;40(21):2321-2332. doi:10.1200/JCO.21.02198.
44. Wang CY, Ginat DT. Preliminary Computed Tomography Radiomics Model for Predicting Pretreatment CD8+ T-Cell Infiltration Status for Primary Head and Neck Squamous Cell Carcinoma. *J Comput Assist Tomogr*. 2021;45(4):629-636. doi:10.1097/RCT.0000000000001149.
45. Tunali I, Hall LO, Napel S, et al. Stability and reproducibility of computed tomography radiomic features extracted from peritumoral regions of lung cancer lesions. *Med Phys*. 2019;46(11):5075-5085. doi:10.1002/mp.13808.
46. Vaidya P, Bera K, Patil PD, et al. Novel, non-invasive imaging approach to identify patients with advanced non-small cell lung cancer at risk of hyperprogressive disease with immune checkpoint blockade. *J Immunother Cancer*. 2020;8(2):e001343. doi:10.1136/jitc-2020-001343.

Tables

- Table 1 Patient characteristics
- Table 2 Performance of the Intratumoral and peritumoral habitat imaging (HI) models across different sequences with corresponding Fusion models
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Figure Legends

Figure 1 Flowchart of patient inclusion.

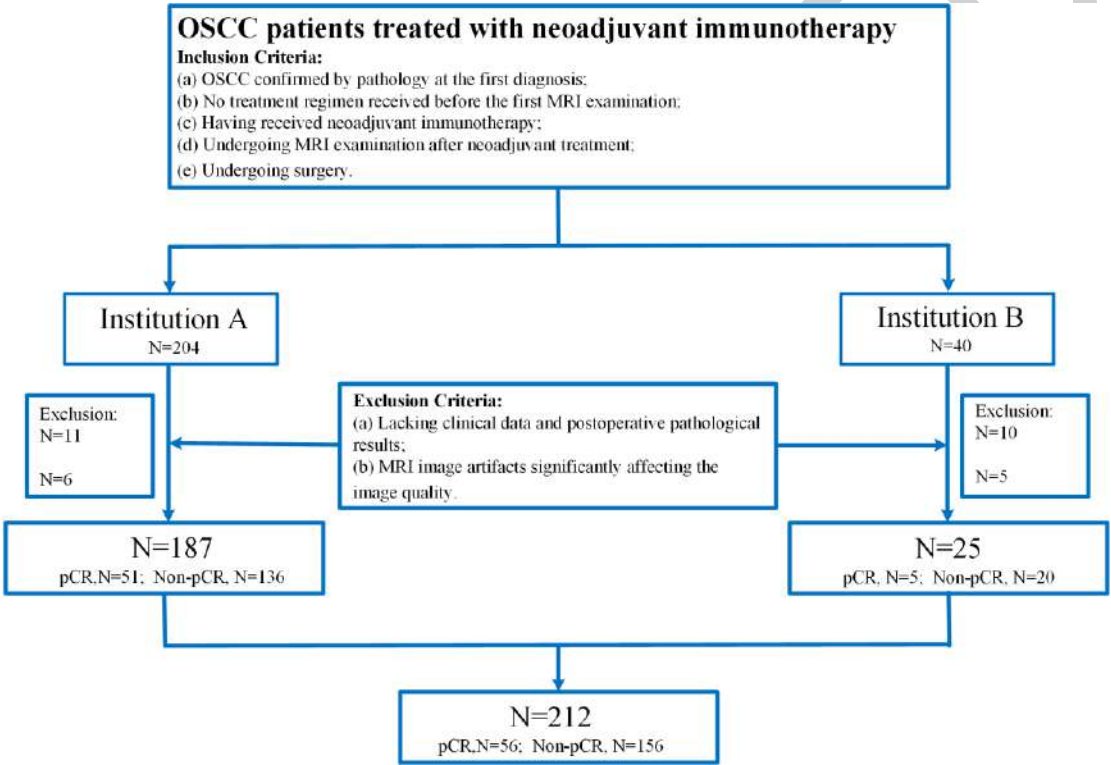


Figure 2 Workflow illustrating radiomics analysis.

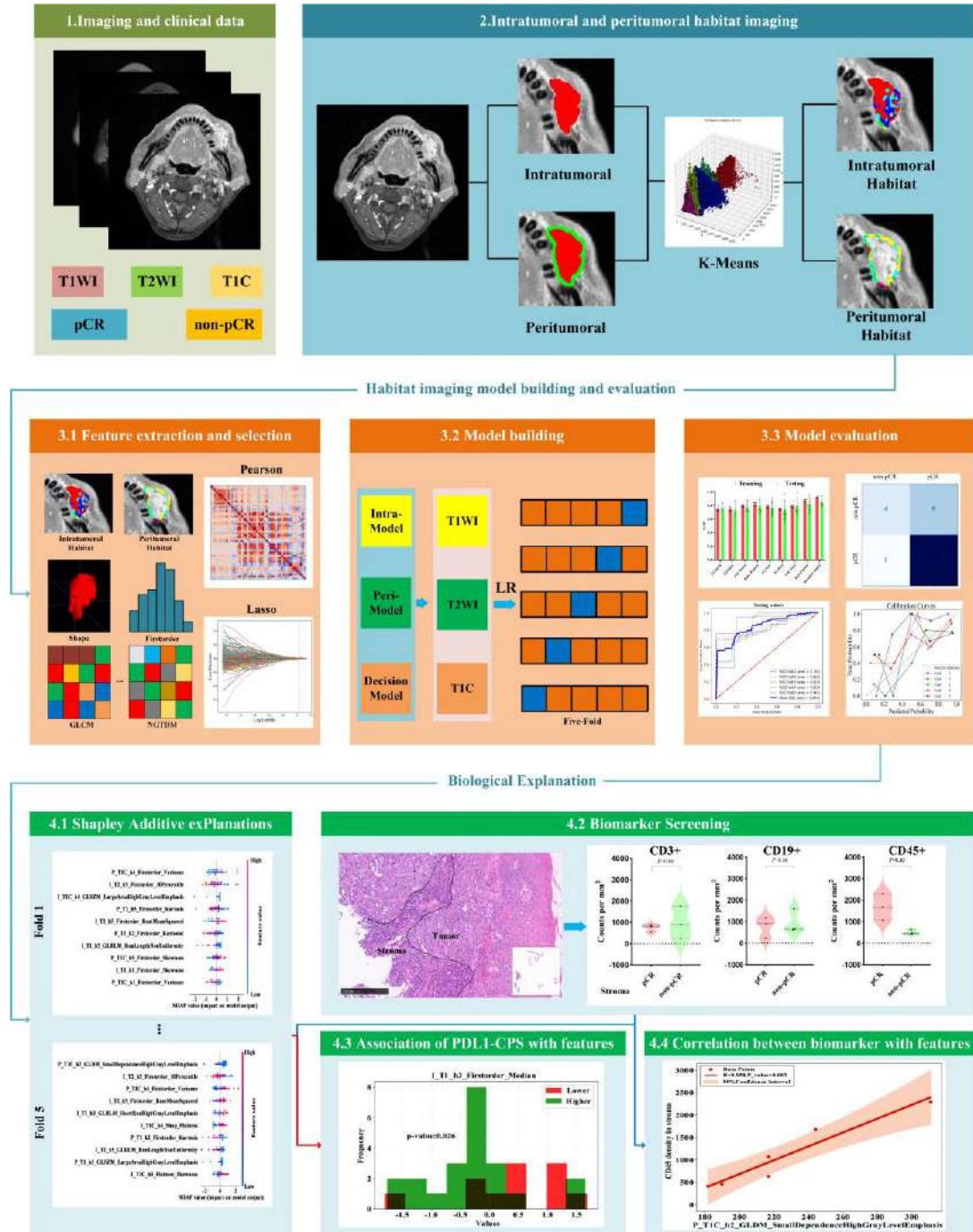


Figure 3 Comparison of different models and feature details in the decision model. (A) Box-and-Whisker plots of AUC values for all models.(B) ROC curves in the training and testing cohorts for the decision model. AUC values are expressed as the mean and standard deviation across the five training and validation sets from the 5-fold

cross-validation. The confusion matrices with heatmaps for the five validation sets of the decision model. The numbers in each colored box represent the percentage of instances between the true and the predicted classes obtained by the decision models. ROC, receiver-operating characteristic; AUC, area under the curve. (C) Details of clinical, peritumoral habitat features (PHF_s) and intratumoral habitat features (IHF_s) included in the decision model. T1, T1-weighted imaging; T1C, contrast-enhanced T1-weighted imaging with fat suppression; T2, axial T2-weighted imaging with fat suppression; Intra, intratumoral; Peri, peritumoral; P, peritumoral; I, intratumoral; h, habitat; GLCM, gray-level co-occurrence matrix; GLRLM, gray-level run-length matrix; GLSZM, gray-level size zone matrix; NGTDM, neighborhood gray-tone difference matrix.

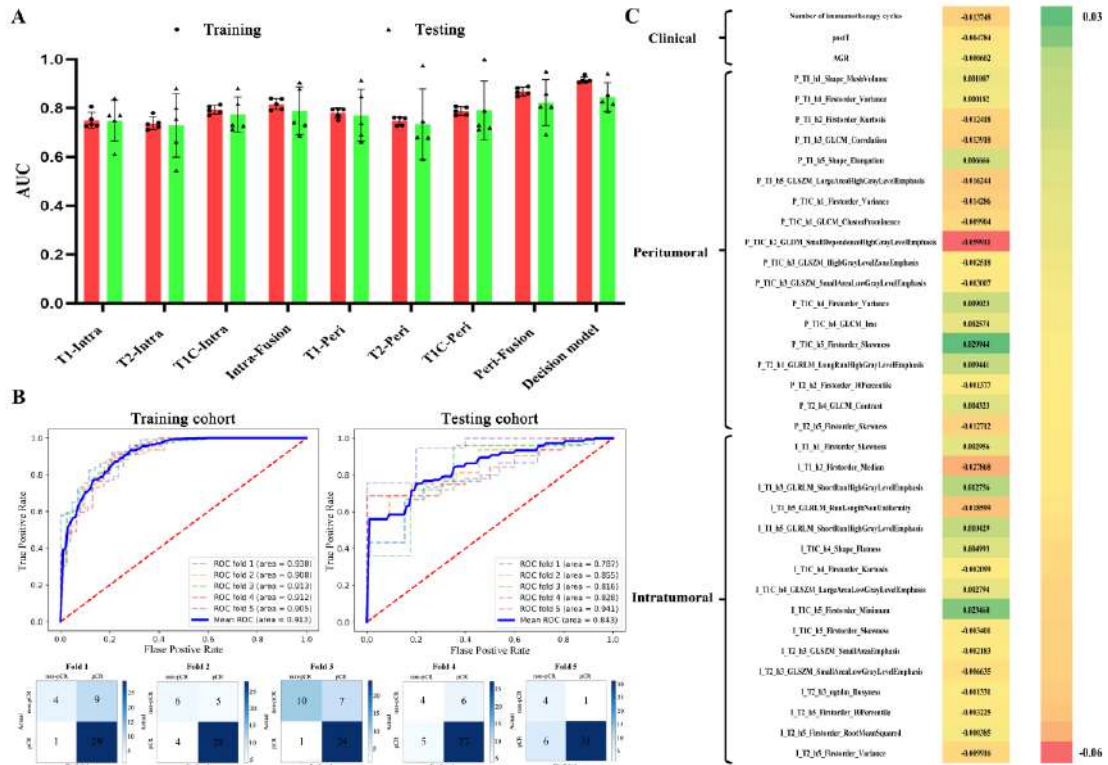


Figure 4 SHAP analysis in the decision model. (A) The SHAP beeswarm plot shows the position or negative effects of each feature on the prediction probability through red and blue colors on a 5-fold training cohort. SHAP summary plot depicting the global

prediction impact of features on the decision model and interaction between features. The importance of features is listed from top to bottom. Each point represents the SHAP value of a patient's feature. Dots to the left of the Y-axis increase the likelihood of pCR, while dots to the right decrease the probability of pCR. SHAP, Shapley additive explanations. Patients (B) (pCR) and (C) (non-pCR) were randomly selected from the decision model. The waterfall plot shows that the impact of the ranking features on the model for a specific patient, where red represents a positive impact and blue represents a negative impact. P, peritumoral; I, intratumoral; h, habitat; GLCM, gray-level co-occurrence matrix, GLRLM, gray-level run-length matrix, GLSZM, gray-level size zone matrix, NGTDM, neighborhood gray-tone difference matrix; pCR, pathologic complete response.

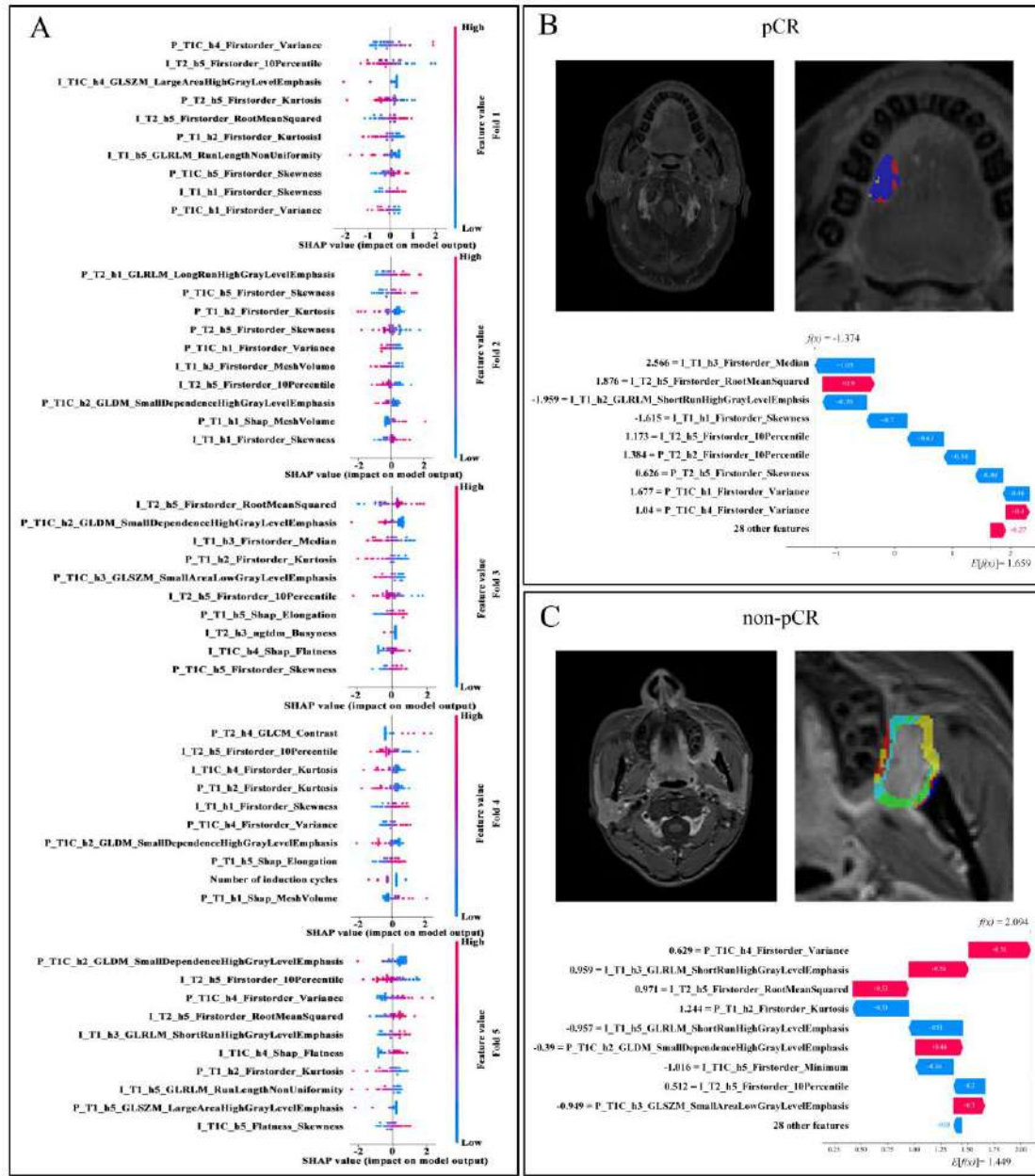
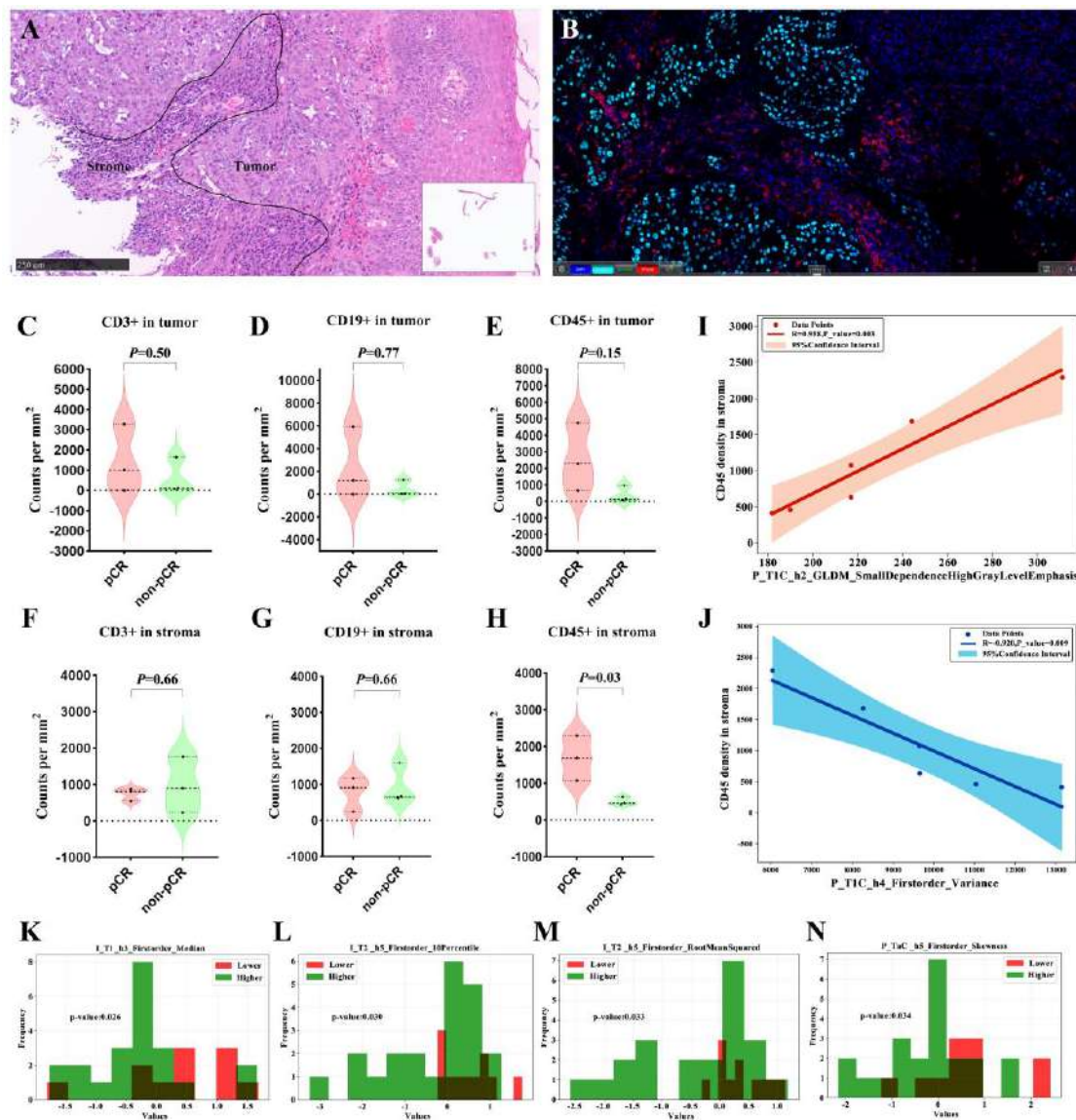


Figure 5 Biological Interpretability Analysis. (A) Pathological specimen section. (B) Hematoxylin - Eosin (H&E) Staining. (C-E) Comparison of CD3+, CD19+ and CD45+ in tumors between pCR and non-pCR groups. (F-H) Comparison of CD3+, CD19+ and CD45+ in stroma between pCR and non-pCR groups. (I) Correlation between P_T1C_h2_GLDM_SmallDependenceHighGrayLevelEmphasis and CD45+ in stroma. (J) Correlation between P_T1C_h4_Firstorder_Variance with CD45+ in stroma. (K-N) Distribution of radiomic features with statistical significance among

patients stratified into different PD-L1 CPS groups. I_T1_h3_Firstorder_Median (K), I_T2_h5_ Firstorder_10Percentile (L), I_T2_h5_ Firstorder_RootMeanSquared (M), P_T1C_h5_ Firstorder_Skewness (N) . Intra, intratumoral; Peri, peritumoral; h, habitat; GLDM, gray-level dependence matrix; T1, T1-weighted imaging; T1C, contrast-enhanced T1-weighted imaging with fat suppression; T2, axial T2-weighted imaging with fat suppression; pCR, pathologic complete response.



Supplementary materials

Table S1 Imaging parameters on different devices

Table S2 Comparison of predictive performance among different machine learning algorithms in intratumoral and corresponding Fusion HI models

Table S3 Comparison of predictive performance among different machine learning algorithms in peritumoral and corresponding Fusion HI models

Table S4 Comparison of predictive performance among different machine learning algorithms in the decision model.

Figure S1 The K-means clustering analysis on intratumoral and peritumoral data from both centers.

Figure S2 Calibration curves of the decision model with five-fold cross-validation in both the training and testing cohorts.

Figure S3 Feature importance ranking based on SHAP (SHapley Additive exPlanations) values in five-fold cross-validation. GLCM, gray-level co-occurrence matrix; GLRLM, gray-level run-length matrix; GLSZM, gray-level size zone matrix; GLDM, gray-level dependence matrix; NGTDM, neighborhood gray-tone difference matrix; P, peritumoral; I, intratumoral; h, habitat.

SDC link : Figures - <http://links.lww.com/JS9/E434>

Table S1 - <http://links.lww.com/JS9/E435>

Table S2 - <http://links.lww.com/JS9/E436>

Table S3 - <http://links.lww.com/JS9/E437>

Table S4 - <http://links.lww.com/JS9/E438>

Table 1. Patient characteristics

	pCR	non-pCR	P
SEX			0.096
Female	6(10.7%)	7(4.5%)	
Male	50(89.3%)	149(95.5%)	
Smoking history			0.579
NO	14(25%)	29(18.6%)	

	Yes	27(48.2%)	84(53.8%)	
	Quit	15(26.8%)	43(27.6%)	
Drinking History				0.139
	No	32(57.1%)	66(42.3%)	
	Yes	16(28.6%)	54(34.6%)	
	Quit	8(14.3%)	36(23.1%)	
preT				0.911
	T1	1(1.8%)	1(0.6%)	
	T2	6(10.7%)	13(8.3%)	
	T3	19(33.9%)	57(36.5%)	
	T4a	20(35.7%)	59(37.8%)	
	T4b	10(17.9%)	26(16.7%)	
preN				0.042*
	N0	24(42.9%)	54(34.6%)	
	N1	8(14.3%)	24(15.4%)	
	N2a	1(1.8%)	3(1.9%)	
	N2b	5(8.9%)	43(27.6%)	
	N2c	4(7.1%)	13(8.3%)	
	N3a	0(0%)	0(0%)	
	N3b	14(25%)	19(12.2%)	
preStage				0.231
	II	1(1.8%)	6(3.8%)	
	III	16(28.6%)	37(23.7%)	
	IVa	19(33.9%)	75(48.1%)	
	IVb	20(35.7%)	38(24.3%)	
Number of induction cycles				0.006*
	1	3(5.4%)	2(1.3%)	
	2	26(46.4%)	85(54.5%)	
	3	23(41.1%)	58(37.2%)	
	4	0(0%)	10(6.4%)	
	5	4(7.1%)	1(0.6%)	
postT				0.006*
	T0	8(14.3%)	23(14.7%)	
	T1	4(7.1%)	19(12.2%)	
	T2	6(10.7%)	38(24.4%)	
	T3	28(50%)	35(22.4%)	
	T4a	6(10.7%)	23(14.7%)	

postN	T4b	4(7.1%)	18(11.5%)	0.702
	N0	33(58.9%)	77(49.4%)	
	N1	8(14.3%)	25(16%)	
	N2a	1(1.8%)	5(3.2%)	
	N2b	7(12.5%)	32(20.5%)	
	N2c	2(3.6%)	7(4.5%)	
	N3a	0(0%)	0(0%)	
	N3b	5(8.9%)	10(6.4%)	
postStage				0.881
	0	16(28.6%)	38(24.4%)	
	I	7(12.5%)	23(14.7%)	
	II	5(8.9%)	14(9.0%)	
	III	13(23.2%)	31(19.9%)	
	IVa	12(21.4%)	34(21.8%)	
	IVb	3(5.4%)	16(10.3%)	
BMI		23.7±4.4	24.1±4.1	0.919
PLT		242.0±78.5	244.2±60.3	0.019*
AGE		54(46.2,59.0)	50(44.0,56.3)	0.146
WBC		6.9(5.8,8.4)	7.1 (6.0,8.2)	0.620
NEU		4.17(3.2,5.4)	4.34(3.5,5.2)	0.549
Monocyte value		0.49(0.4,0.6)	0.6(0.4,0.7)	0.135
MON		6.7(5.6,7.5)	7.2(5.9,8.4)	0.046*
HGB		150(137.8,160.3)	150(140.0,158.0)	0.906
PDW		11.6(10.8,13.3)	11.6(10.4,12.8)	0.577
ALT		23(16.9,33.0)	24.0(17.1,39.5)	0.458
AST		20.3(16.3,28.1)	21.45(17.4,29.0)	0.369
ALB		43.3(41.0,45.0)	43.2(41.1,45.5)	0.802
GLO		28.2 (25.7,32.4)	27.5(25.0,30.8)	0.214
AGR		1.5 (1.3,1.7)	1.6 (1.4,1.7)	0.250

*Statistically significant. BMI:body mass index; PLT: platelet; WBC: white blood cell count; NEU: neutropenia; MON: monocyte; HGB: hemoglobin; PDW: platelet distribution width; ALT: alanine aminotransferase; AST: Aspartate transaminase; ALB: albumin; GLO:globulin; AGR: albumin-globulin ratio.

Table 2 Performance of the Intratumoral and peritumoral habitat imaging (HI) models across different sequences with corresponding Fusion models.

	AUC	Accurac	Recal	Precisio	F1-scor	PPV	NPV
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			y	l	n	e		
T1-Intra	Trai	0.75					0.77	0.68
	n	0	0.765	0.966	0.772	0.858	2	4
	Test	0.74					0.77	0.64
		8	0.759	0.968	0.770	0.852	0	0
T1C-Intra	Trai	0.79					0.80	0.65
	n	3	0.781	0.930	0.802	0.861	2	6
	Test	0.77					0.76	0.46
		4	0.722	0.900	0.769	0.824	9	5
T2-Intra	Trai	0.73					0.76	0.58
	n	8	0.738	0.948	0.767	0.848	7	5
	Test	0.72					0.75	0.40
		9	0.708	0.914	0.752	0.819	2	4
Intra-Fusio	Trai	0.81					0.81	0.71
	n	7	0.801	0.942	0.816	0.874	6	5
	Test	0.78					0.80	0.66
		9	0.774	0.926	0.803	0.789	3	3
T1-Peri	Trai	0.78					0.79	0.73
	n	1	0.789	0.956	0.797	0.869	7	3
	Test	0.77					0.77	0.55
		1	0.741	0.926	0.773	0.837	3	1
T1C-Peri	Trai	0.78					0.79	0.68
	n	8	0.777	0.944	0.791	0.861	1	7
	Test	0.79					0.78	0.55
		0	0.736	0.901	0.782	0.828	2	0
T2-Peri	Trai	0.74					0.77	0.64
	n	8	0.763	0.950	0.777	0.855	7	2
	Test	0.73					0.75	0.49
		4	0.731	0.928	0.759	0.831	9	0
Peri-Fusion	Trai	0.86					0.83	0.71
	n	9	0.810	0.930	0.831	0.877	1	8
	Test	0.82					0.78	0.49
		2	0.727	0.902	0.783	0.827	3	3

T1: T1-weighted imaging; T1:T1-weighted imaging with fat suppression; T2:axial T2-weighted imaging with fat suppression; AUC:Area Under Curve;PPV: PositivePredictive Value; NPV: Negative predictive value; Intra: intratumoral; Peri: peritumoral.

Table 3 Performance of the decision model with five-fold cross-validation.

Fold	Cohort	AUC	Accuracy	Recall	Precision	F1-score	PPV	NPV
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Fold 1	Train	0.938	0.888	0.96	0.896	0.927	0.896	0.853
	Test	0.787	0.767	0.967	0.763	0.853	0.763	0.800
Fold 2	Train	0.908	0.852	0.952	0.861	0.904	0.861	0.812
	Test	0.855	0.791	0.875	0.848	0.812	0.848	0.600
Fold 3	Train	0.913	0.888	0.969	0.894	0.93	0.894	0.857
	Test	0.816	0.810	0.96	0.774	0.857	0.774	0.909
Fold 4	Train	0.912	0.888	0.96	0.895	0.926	0.895	0.865
	Test	0.816	0.81	0.96	0.774	0.857	0.774	0.909
Fold 5	Train	0.905	0.859	0.933	0.874	0.902	0.874	0.814
	Test	0.941	0.833	0.838	0.969	0.814	0.969	0.400
Mean	Train	0.913	0.875	0.955	0.884	0.918	0.884	0.840
	Test	0.843	0.788	0.897	0.835	0.840	0.835	0.631

PPV: Positive Predictive Value; NPV: Negative predictive value; AUC: Area Under Curve.

Table 4. Associations between the key radiomic features selected by SHAP and both

Radiomics Features	CPS		CD45+	
	t	p	r	p
P_T1_h2_Firstorder_Kurtosis	-0.357	0.723	0.406	0.423
P_T1C_h2_GLDM_SmallDependence HighGrayLevelEmphasis	-0.339	0.736	0.958	0.003*
P_T1C_h4_Firstorder_Variance	-0.649	0.521	-0.919	0.009*
P_T1C_h5_Firstorder_Skewness	2.222	0.033*	-0.164	0.754
P_T2_h1_GLRLM_LongRunHigh GrayLevelEmphasis	-0.813	0.422	0.343	0.504
P_T2_h4_GLCM_Contrast	-0.304	0.762	-0.224	0.668
I_T1_h3_Firstorder_Median	2.344	0.025*	-0.173	0.742
I_T1C_h4_Firderorder_Kurtosis	0.201	0.841	-0.281	0.588

PD-L1 CPS and CD45+ in the stroma.

I_T1C_h4_GLSZM_LargeAreaLowGray LevelEmphasis	2.027	0.069	-0.102	0.846
I_T2_h5_Firstorder_10Percentile	2.275	0.029*	0.131	0.804
I_T2_h5_Firstorder_RootMean Squared	2.233	0.032*	0.065	0.901

I:intratumoral; P: peritumoral; T1:T1-weighted imaging; T1C:T1-weighted imaging with fat suppression; T2:axial T2-weighted imaging with fat suppression; h: habitat. * Statistically significant.

ACCEPTED

Hightlights

In a retrospective analysis of 212 OSCC patients undergoing neoadjuvant chemoimmunotherapy, multi-sequence MRI-derived intratumoral and peritumoral habitat imaging demonstrated robust predictive value for pCR.

The integrated decision model combining peritumoral habitat features (PHFs), intratumoral habitat features (IHF), and clinical characteristics achieved superior predictive performance (AUC 0.913/0.843 in training/testing cohorts), outperforming single-sequence or single-region models.

SHAP-based visualization identified critical habitat features were associated with PD-L1 CPS and CD45+ in the stroma which was correlated with pCR, providing biological interpretability for personalized treatment optimization.